USN | | | | | | | | | | |

[22ESME11]

FIRST SEMESTER B.Tech EXAMINATION, DECEMBER -2022

ENGINEERING MECHANICS

Time: 3 Hours **Maximum Marks: 100**

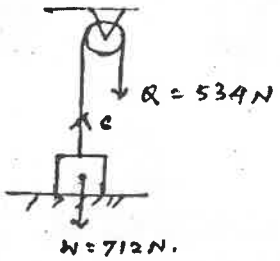
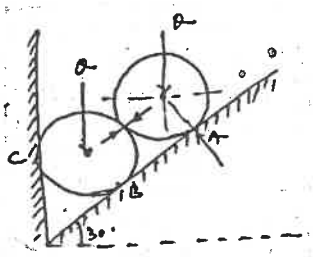
Instructions: Missing data may be suitably assumed.

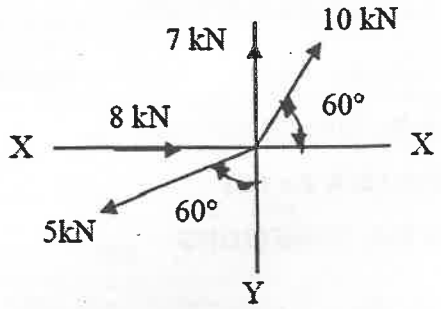
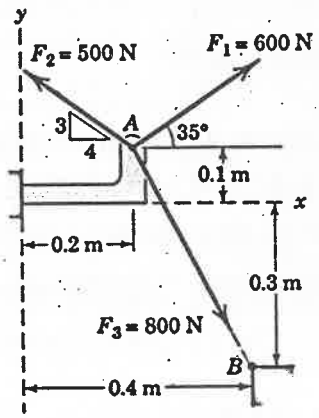
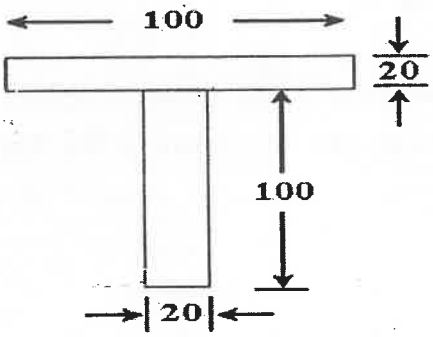
PART-A (5 X 2=10)

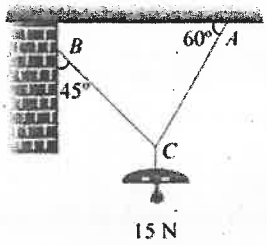
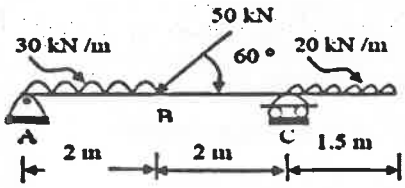
ANSWER ALL QUESTIONS

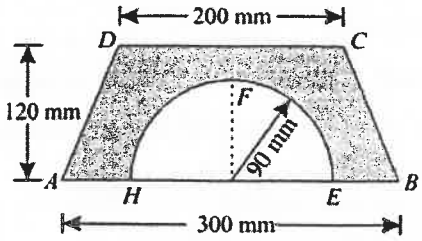
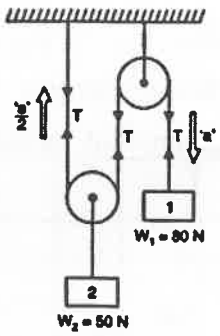
1	a.	Explain law of parallelogram of forces.	L- 2	CO1	02
	b.	Elucidate free body diagram with example.	L- 2	CO2	02
	c.	Explain any two system of forces.	L- 2	CO3	02
	d.	Define moment and moment of inertia.	L- 1	CO4	02
	e.	Classify the types of friction.	L- 2	CO5	02

PART-B (6 X 5=30)

2.	a.	A man of weight W = 712 N holds one end of a rope that passes over a pulley vertically above his head and to the other end of which is attached a weight Q = 534 N. Find the force with which the man's feet press against the floor shown in the Fig.2a  **Fig.2a**	L- 3	CO1	05
		OR			
	b.	Two identical rollers each of weight Q = 445 N are supported by an inclined plane and a vertical wall as shown in the Fig.2b Assuming smooth surfaces, find the reactions induced at the points of support A, B and C  **Fig.2b**	L- 3	CO1	05

3.	a.	Find resultant of a force system shown in Fig.3a  **Fig.3a**	L- 3	CO2	**05**
		OR			
	b.	Determine the x and y scalar components of F1, F2, and F3 acting at point A of the bracket shown in the Fig. 3b  **Fig. 3b**	L- 3	CO2	**05**
4.	a.	Find the centroid of the T-section as shown in Fig. 4a from the bottom (dimensions are in mm)  **Fig. 4a**	L- 3	CO3	**05**
		OR			
	b.	Find the centroid of an unequal angle section 100 mm × 80 mm × 20 mm.	L- 3	CO3	**05**

5.	a.	Find the moment of inertia of a circular section.	L- 2	CO4	05
		OR			
	b.	State and explain parallel axis theorem.	L- 2	CO4	05
6.	a.	A car of mass 1000 kg moving with a velocity of 10m/s is acted upon by a forward force of 1000 N due to engine and retarding force of 500 N due to friction. Find it's velocity after 10s?	L- 3	CO5	05
		OR			
	b.	Explain law of friction and its advantages.	L- 2	CO5	05
7.	a.	Find the magnitude of the two forces, such that if they act at right angles, their resultant is 10 N. But if they act at 60°, their resultant is 13 N.	L- 3	CO6	05
		OR			
	b.	Elucidate the terms centre of gravity and centroid.	L- 2	CO6	05
		PART-C (6 X 10=60)			
8.	a.	An electric light fixture weighting 15 N hangs from a point C, by two strings AC and BC. The string AC is inclined at 60° to the horizontal and BC at 45° to the horizontal as shown in Fig. 8a Using Lami's theorem, or otherwise, determine the force in the strings AC  **Fig. 8a**	L- 3	CO1	10
		OR			
	b.	Two equal heavy spheres of 50 mm radius are in equilibrium within a smooth cup of 150 mm radius. Show that the reaction between the cup of one sphere is double than that between the two spheres.	L- 3	CO1	10
9.	a.	Explain system of force with neat sketch	L- 2	CO2	10
		OR			
	b.	Find out the support reactions for the beam shown in the Fig. 9b  **Fig. 9b**	L- 4	CO2	10

10.	**a.**	A semicircle of 90 mm radius is cut out from a trapezium as shown in Fig. 10a.Find the position of the centre of gravity.  **Fig. 10a**	L- 3	CO3	**10**
		OR			
	b.	An I-section has the following dimensions in mm units: Bottom flange = 300 × 100 Top flange = 150 × 50 Web = 300 × 50 Determine mathematically the position of centre of gravity of the section.	L–3	CO3	**10**
11.	**a.**	Deduce the moment of moment of inertia of a rectangular section and hollow rectangular section.	L–2	CO4	**10**
		OR			
	b.	Find the moment of inertia of a T-section with flange as 150 mm × 50 mm and web as 150 mm × 50 mm about X-X and Y-Y axes through the centre of gravity of the section.	L- 2	CO4	**10**
12.	**a.**	A system of frictionless pulleys carries two weights hung by inextensible cords as shown in Fig. 12 a Find : **i.** The acceleration of the weights and tension in the cords. **ii.** The velocity and displacement of weight '1' after 5 seconds from start if the system is released from rest.  **Fig. 12a**	L–3	CO5	**10**
		OR			
	b.	Find the acceleration of a solid body A of weight 8 N, when it is being pulled by another body of weight 6 N along a smooth horizontal plane as shown in Fig. 12b	L- 3	CO5	**10**

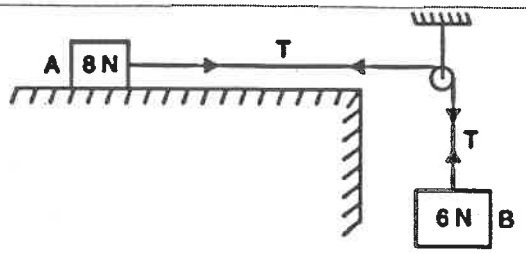


Fig. 12b

13.	**a.**	Two blocks shown in Fig. 13a. have weights A = 8 N and B = 10 N and co-efficient of friction between the block A and horizontal plane is 0.2 If the system is released, from rest and the block A falls through a vertical distance of 1.5 m, what is the velocity acquired? Neglect the friction in the pulley and extension of the string.	L- 3	CO6	**10**

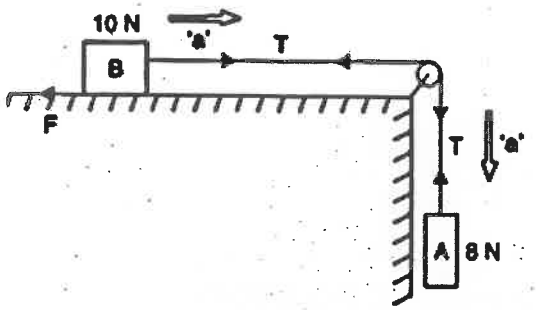


Fig. 13a

OR

	b.	Block A weighing 1000N rests over block B which weighs 2000N as shown in Fig. 13b. Block A is tied to wall with a horizontal string. If the coefficient of friction between blocks A and B is 0.25 and between B and floor is 0.34, what should be the value of P to move the block (B), if **i.** P is horizontal. **ii.** P acts at 30° upwards to horizontal.	L- 4	CO6	**10**

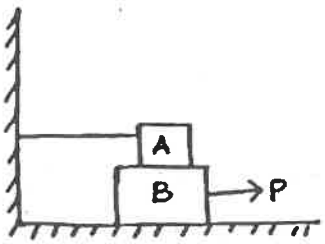


Fig. 13b